



Project Report

Comparison of Axial, Radial, and Mixed Flow Pumps

Fluid Mechanics Project Report - Group 5

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1. Abstract:

This project report provides a comprehensive comparative analysis of rotodynamic fluid machinery, specifically focusing on radial, axial, and mixed flow pumps. Pumps are critical infrastructure components across almost every engineering domain, from power generation and automotive cooling systems to civil flood control and marine ballast management. The primary objective of this report is to evaluate how the fundamental kinematic principles of fluid momentum transfer dictate the operational characteristics, efficiency curves, and industrial applications of these three primary pump categories. By analyzing the mechanisms of energy transfer—whether it be purely centrifugal force, aerodynamic lift, or a hybridized approach—this study analyses the direct correlation between impeller geometry and performance outcomes [1, 3].

Furthermore, this report delves deeply into the concept of specific speed (N_s), a vital dimensionless parameter that serves as the ultimate index for pump selection. The relationship between head (H), discharge (Q), and specific speed will be explored to demonstrate why radial pumps are suited for high-head, low-flow scenarios, why axial pumps dominate in high-volume, low-head environments, and how mixed flow pumps bridge the operational gap [6]. Finally, an assessment of performance curves (head-discharge and power-discharge) will highlight critical operational safety parameters, such as the danger of shut-off head in axial pumps versus radial pumps [6, 7].

2. Introduction:

Background and Context: The transportation of fluids is a foundational requirement in mechanical, civil, and chemical engineering. Fluid machinery designed to impart energy to a fluid, thereby increasing its pressure or velocity, are broadly classified as pumps. Among the diverse array of pump designs, they are universally categorized into two main branches: positive displacement pumps and dynamic pumps [1, 4]. While positive displacement pumps physically trap and force a fixed volume of fluid through a system, dynamic pumps operate continuously by transferring kinetic energy from a rotating impeller to the fluid, which is subsequently converted into pressure energy via a stationary casing or diffuser [3].

Dynamic pumps are further classified based on the principal direction of fluid flow as it passes through the rotating impeller relative to the axis of rotation. This geometric classification yields three distinct categories: radial flow (commonly known as centrifugal), axial flow (propeller), and mixed flow pumps. The direction of this flow is not merely a structural choice; it fundamentally dictates the mechanism of energy transfer.

Objectives of the Study: The purpose of this extensive review is multifold:

1. To rigorously define the operating principles and internal kinematics of radial, axial, and mixed flow pumps.
2. To mathematically and conceptually analyze the concept of specific speed and its role in pump standardization.
3. To compare the characteristic performance curves of each pump type, exploring how efficiency, required power, and generated head fluctuate with varying volumetric flow rates.
4. To map these theoretical characteristics to real-world industrial and engineering applications, providing a practical framework for pump selection.

Understanding the nuances between these three pump types is critical for engineers. Improper pump selection can lead to catastrophic system failures, severe energetic inefficiencies, and phenomena such as cavitation, which can rapidly destroy an impeller through localized micro-implosions [7].

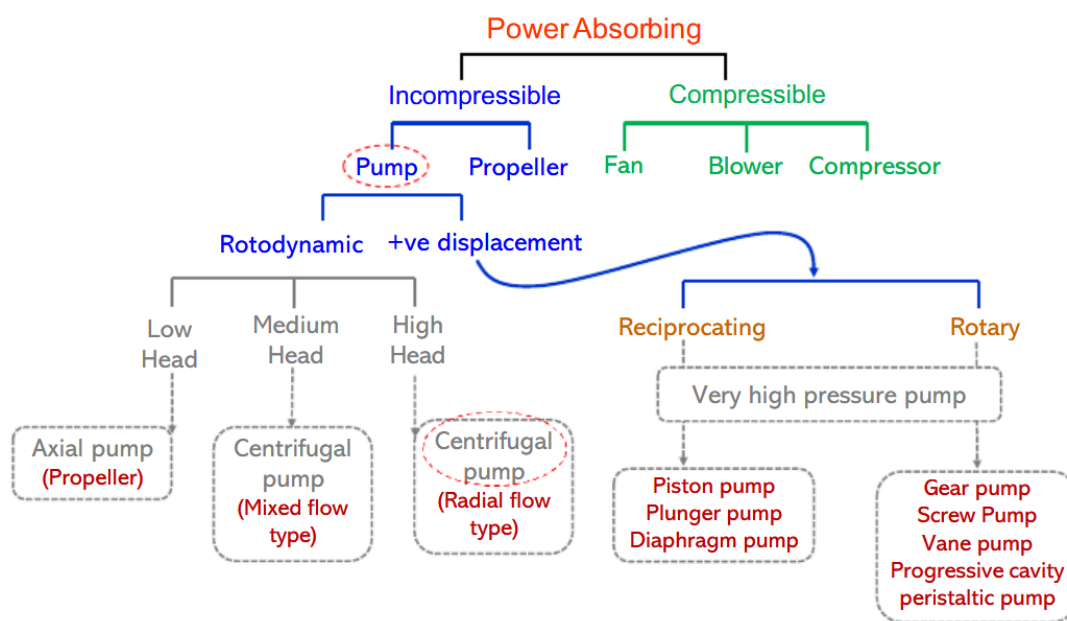


Figure 1: Classification of Rotodynamic Pumps based on Flow Direction

3. Radial Flow (Centrifugal) Pumps:

Operating Principle and Energy Transfer: Radial flow pumps, universally referred to as centrifugal pumps, universally represents the category of dynamic fluid machinery in modern engineering. The fundamental operating principle relies exclusively on the generation of centrifugal force to impart kinetic energy to the working fluid. In this configuration, the fluid enters the pump casing axially through the center, known as the ‘eye’ of the impeller. Upon entry, the fluid is engaged by the rotating blades or vanes. As the impeller spins—driven by a prime mover such as an electric motor—the fluid is forced radially outward towards the periphery of the impeller due to centrifugal acceleration.

The kinetic energy transferred to the fluid results in a high-velocity discharge at the outer diameter of the impeller. However, high kinetic energy is rarely the desired output for fluid transport, high static pressure is required to overcome system resistance and elevation changes. Therefore, the fluid enters a stationary volute casing or a diffuser ring. The volute features a progressively increasing cross-sectional area, which decelerates the fluid flow. According to Bernoulli’s principle, this controlled reduction in kinetic energy translates directly into an increase in static pressure energy [1, 4].

Impeller Design and Velocity Triangles: The design of the radial impeller vanes plays a crucial role in the performance characteristics. Vanes are categorized into three primary configurations based on their exit angle: backward-curved, radial, and forward-curved. Backward-curved vanes are the industry standard for most centrifugal pumps. An analysis of the velocity triangles at the impeller exit reveals that backward-curved vanes offer a stable head-discharge characteristic and possess a self-limiting power curve [2]. This means the prime mover is less likely to overload at exceedingly high flow rates. Forward-curved vanes can deliver a higher absolute head for a given size and rotational speed, but they suffer from unstable performance curves and power requirements that rise sharply with increased flow, potentially causing motor burnout [5].

Characteristics and Applications: Radial flow pumps are inherently designed to generate high head (pressure) at the expense of massive volumetric flow rates. Because the fluid travels

from a small inner radius to a much larger outer radius, the centrifugal head generated is substantial. Consequently, these pumps possess a low specific speed (N_s) [2, 6].

Industrial applications for radial pumps are vast and include:

- **Boiler Feed Systems:** Where water must be injected into steam boilers against immense internal pressures.
- **Reverse Osmosis:** Supplying the extreme pressures required to force water through semi-permeable membranes.
- **Automotive Cooling:** Internal combustion engine water pumps are typically compact radial pumps.
- **Deep Well Extraction:** Multi-stage radial pumps are often submerged to push water up from significant depths.

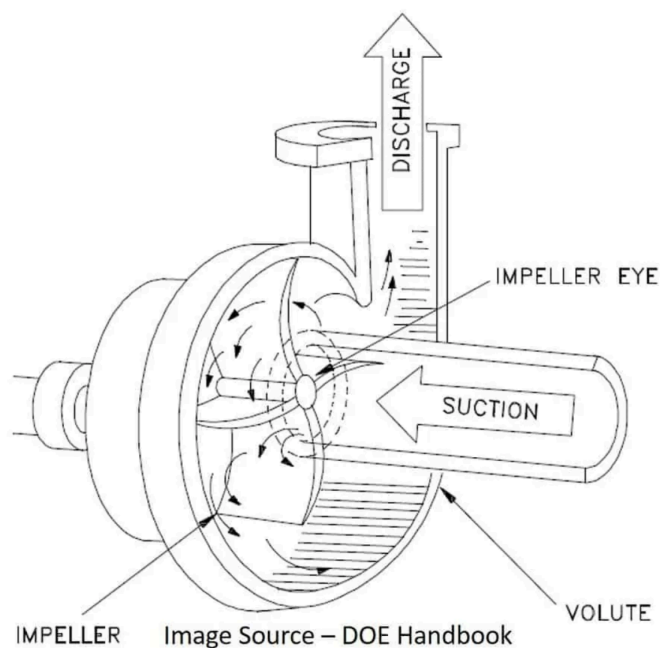


Figure 2: Cross-section of a Radial Flow Pump

4. Axial Flow Pumps:

Operating Principle and Aerodynamic Lift: In stark contrast to radial pumps, axial flow pumps (frequently termed propeller pumps) do not rely on centrifugal force for energy transfer. Instead, the fluid enters and exits the impeller parallel to the axis of rotation. The pumping action is generated entirely by the aerodynamic lift created by the propeller blades acting on the fluid [2, 5].

The impeller of an axial flow pump closely resembles the propeller of a ship or an airplane. As the blades cut through the fluid, the physical camber and angle of attack of the blade profiles create a pressure differential—lower pressure on the leading face and higher pressure on the trailing face. This pressure differential essentially pushes the fluid axially down the pipe. Because the radial distance of the fluid from the axis of rotation does not change significantly from inlet to outlet, there is zero centrifugal contribution to the pressure head. The stationary casing surrounding the propeller often contains guide vanes (stators) to remove the swirl or rotational velocity imparted by the impeller, straightening the flow and recovering additional pressure.

Characteristics and The Power Paradox: Axial flow pumps are characterized by their ability to move massive volumetric flow rates against very low resistance or head. This translates to an exceptionally high specific speed (N_s). Because the flow path is unobstructed and relatively straight, hydraulic losses due to sharp directional changes are minimized, allowing for high efficiency at peak flow conditions [4].

A critical engineering consideration for axial flow pumps lies in their power curve. Unlike radial pumps, where the power requirement generally increases as flow rate increases, an axial pump requires its maximum power when the flow is zero (shut-off condition). If a system valve is closed while the pump is running, the steep angle of the blades stalling against the stationary fluid causes a massive spike in torque and power consumption. For this reason, axial pumps are almost never started against a closed discharge valve, whereas radial pumps routinely are [6].

Applications: The unique high-flow, low-head characteristics of axial pumps make them indispensable for specific, large-scale engineering applications:

- **Flood Control and Drainage:** Moving massive volumes of storm water over low levees or barriers.
- **Irrigation Systems:** Drawing water from rivers into low-lying agricultural canals.
- **Ship Ballast and Waterjets:** Rapidly transferring water between ballast tanks to control a ship's draft and trim.
- **Evaporator and Crystallizer Circulation:** Moving large volumes of viscous fluids with minimal pressure changes in chemical processing.

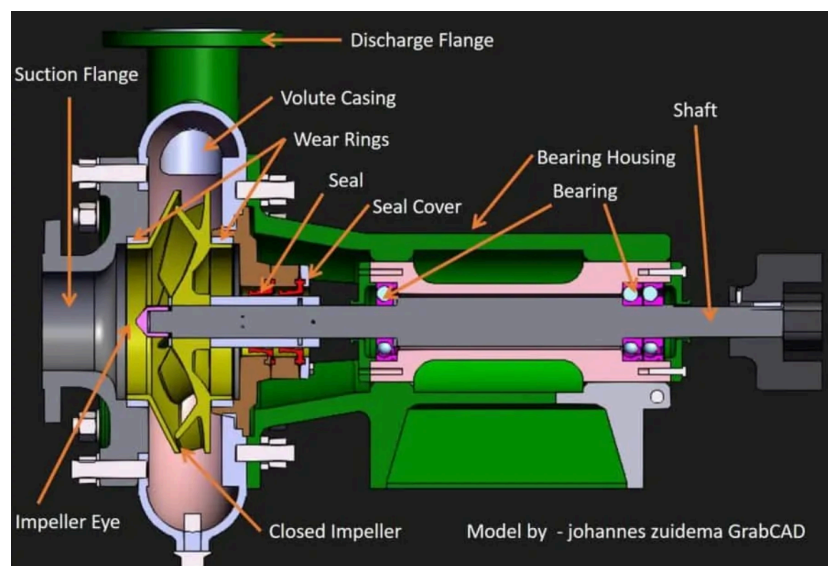


Figure 3: Cross-section of an Axial Flow Pump

5. Mixed Flow Pumps:

Operating Principle: The Hybrid Approach: Mixed flow pumps represent an engineering compromise designed to operate in the performance gap left by pure radial and pure axial pumps. As the name implies, the mechanics of a mixed flow pump utilize a combination of both centrifugal force and aerodynamic lift to impart energy to the fluid.

In a mixed flow impeller, the fluid enters axially along the center of rotation but is discharged at a diagonal angle—typically between 0° and 90° relative to the axis [5]. The physical geometry of the impeller is distinctly recognizable; the blades are highly twisted, resembling a stretched-out centrifugal impeller or a deeply concaved propeller. As the fluid travels through the impeller, it moves outward to a larger radius (generating centrifugal head) while simultaneously being pushed forward by the pitch of the twisted vanes (generating axial lift).

The casing surrounding a mixed flow impeller can vary. In some designs, a traditional volute is utilized, similar to a radial pump. In others, particularly in vertical turbine applications, the pump utilizes a bowl assembly with diffusion vanes. The diffuser serves to gently redirect the diagonal flow back to a purely axial direction for the next stage or for discharge up a column pipe.

Characteristics and Versatility: Because it employs a dual-mechanism for energy transfer, the mixed flow pump offers a highly versatile performance profile. It generates a medium-level pressure head—less than a comparably sized radial pump but significantly more than an axial pump. Concurrently, it can handle large volumetric flow rates, less than an axial pump but much greater than a radial pump. This places the mixed flow pump solidly in the intermediate specific speed (N_s) range [6, 7].

Applications: The balanced metrics of mixed flow pumps make them highly sought after in scenarios where neither extreme high pressure nor extreme volume alone will suffice:

- **Municipal Water Supply:** Pumping large volumes of treated water into elevated city reservoirs.
- **Cooling Water for Power Plants:** Supplying massive amounts of water through condenser networks in thermal and nuclear power stations.

- **Sewage and Wastewater Treatment:** Handling large flows with suspended solids where pure axial pumps might clog and pure radial pumps lack the volume capacity.

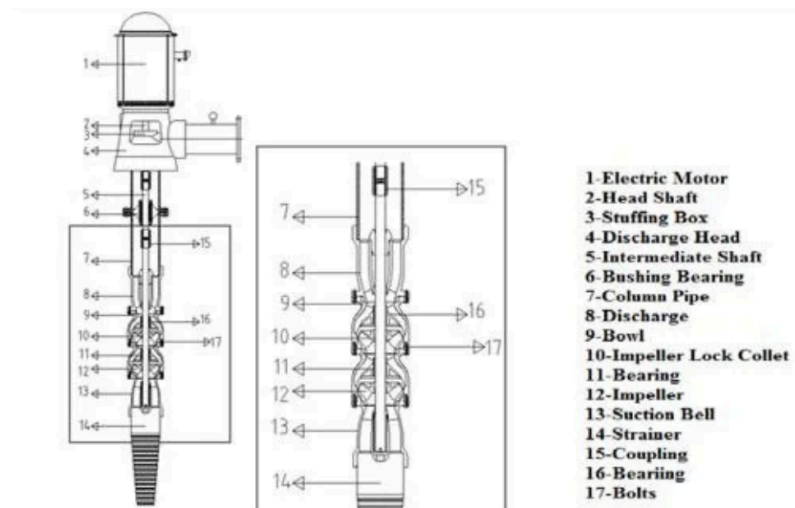


Figure 4: Cross-section of a Mixed Flow Pump

6. Comparative Analysis & Specific Speed:

Specific Speed (N_s) as a Classification Index: The most rigorous mathematical method for comparing these diverse pump geometries is through the parameter of specific speed. Specific speed is a dimensionless number that serves as an index correlating pump capacity, head, and rotational speed at the point of maximum efficiency (Best Efficiency Point - BEP). It determines the optimal geometric shape of the impeller regardless of its physical size.

The equation for specific speed is defined as [2, 6]:

$$N_s = \frac{N\sqrt{Q}}{H^{\frac{3}{4}}}$$

Where:

- N = Pump rotational speed (revolutions per minute)
- Q = Volumetric flow rate ($\frac{m^3}{s}$ or gallons per minute)
- H = Total generated head (m or feet)

Interpreting the equation mathematically reveals the physical realities of the pumps:

- To achieve a high H and low Q , the denominator becomes large and the numerator small, resulting in a Low N_s . This requires a large outer diameter relative to the eye, dictating a Radial design.
- To achieve a massive Q with very low H , the numerator grows while the denominator shrinks, resulting in a High N_s . This dictates an Axial propeller design.
- Moderate values of both yield a Medium N_s , requiring a Mixed flow geometry.

Table 1: Comprehensive Comparison Matrix

Parameter	Radial Flow (Centrifugal)	Mixed Flow	Axial Flow (Propeller)
Flow Direction at Exit	Perpendicular (90°) to axis	Diagonal to axis	Parallel (0°) to axis
Head Generation (H)	High	Medium	Low
Discharge Capacity (Q)	Low to Medium	Medium to High	Very High

Parameter	Radial Flow (Centrifugal)	Mixed Flow	Axial Flow (Propeller)
Specific Speed (N_s US Units)	Low (500 – 4000)	Medium (4000 – 10000)	High (> 10000)
Energy Transfer Mechanism	Pure Centrifugal Force	Centrifugal + Aerodynamic Lift	Pure Aerodynamic Lift
Power at Shut-off Head	Minimum (Safe to start)	Moderate	Maximum (Danger of stall/burnout)

Analysis of Characteristic Performance Curves: Comparing the operational curves (Head, Power, and Efficiency plotted against Flow Rate) provides the ultimate practical insight into pump selection.

1. **Head vs. Discharge (H-Q Curve):** A radial pump typically exhibits a relatively flat or gently drooping curve, maintaining pressure well across varying flows. An axial pump features a drastically steep H-Q curve; head drops off exponentially as flow increases.
2. **Power vs. Discharge (P-Q Curve):** As previously noted, the power curve for a radial pump rises from a minimum at zero flow to a maximum near the Best Efficiency Point (BEP). Conversely, an axial pump's power curve plummets from a dangerous maximum at zero flow to a minimum at high flow. Mixed flow pumps feature a relatively flat power curve, making them highly resilient to system variations without risking motor overload.
3. **Efficiency:** All pumps operate along a parabolic efficiency curve. Selecting the correct pump type ensures that the system's operational requirements intersect the pump's curve exactly at the BEP, minimizing energetic waste and mechanical vibration [4, 7].

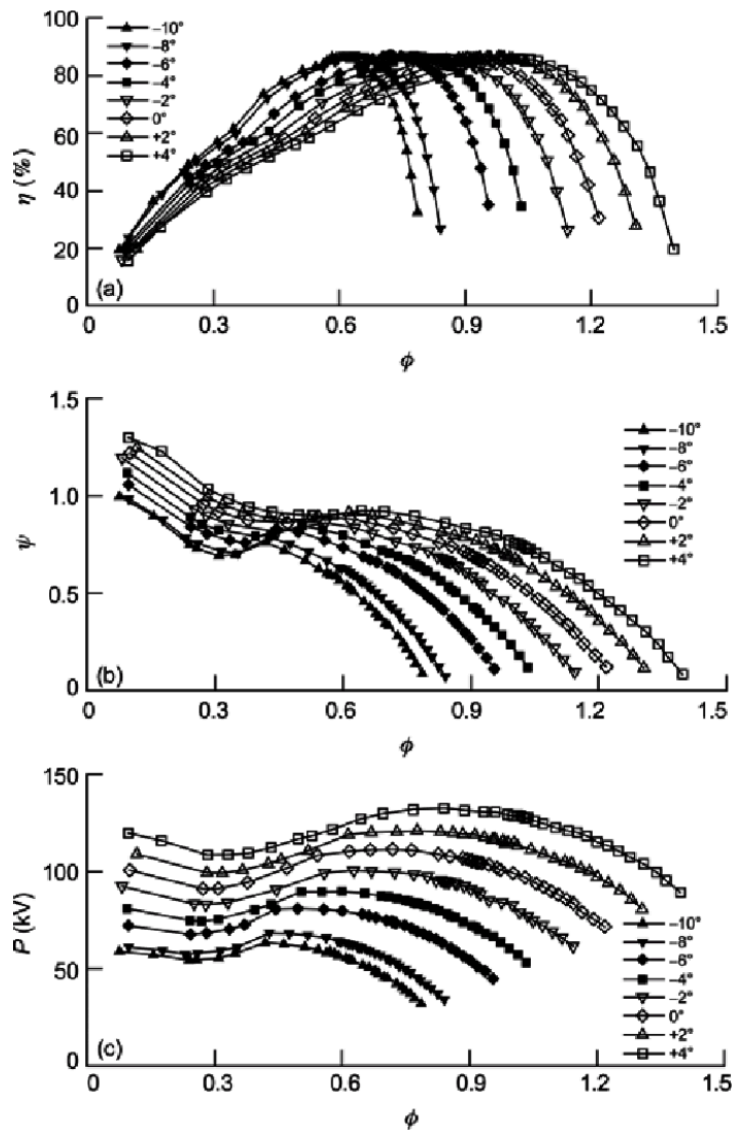


Figure 5: Comparative Performance Curves (Head, Power, and Efficiency vs. Flow Rate)

7. Conclusion:

The selection between radial, mixed, and axial flow pumps is never a matter of arbitrary preference, it is fundamentally dictated by the strict hydrodynamic requirements of the engineering system in question. Radial flow pumps harness centrifugal acceleration to conquer extreme pressure differentials, making them the backbone of industrial processing, boiler systems, and fluid extraction. Axial flow pumps rely on aerodynamic lift to move astronomical volumes of fluid against negligible resistance, serving as the unsung heroes of municipal flood control and large-scale agricultural irrigation. Mixed flow pumps intelligently utilize both physical mechanisms, providing a highly adaptable, robust solution for intermediate industrial demands.

Ultimately, the mastery of pump selection rests upon the profound understanding of specific speed (N_s) and characteristic performance curves. By correctly calculating the system's required head and discharge, and matching it to the corresponding specific speed, engineers ensure that the chosen machinery operates precisely at its Best Efficiency Point [2, 6]. This careful calibration not only guarantees optimal fluid transport but actively mitigates the risks of catastrophic mechanical failure, motor overloading, and cavitation degradation, thereby ensuring the longevity and safety of the entire fluid system [6, 7].

8. References:

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